

Probability
Chapter 2 - Module 6

Additional Remarks on Chapter 2.

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Example 1. Let $g(x) = \frac{1}{1+x^2}$. Let's normalize this so that it is a density function.

$$\begin{aligned}
 \int_{-\infty}^{\infty} \frac{1}{1+x^2} dx &= \lim_{a \rightarrow \infty} \int_{-a}^a \frac{1}{1+x^2} dx \\
 &= \lim_{a \rightarrow \infty} \arctan x \Big|_{-a}^a \\
 &= \lim_{a \rightarrow \infty} 2 \arctan a \\
 &= 2 \pi/2 = \pi.
 \end{aligned}$$

So

$$f(x) = \frac{1}{\pi} \frac{1}{1+x^2}$$

is a pdf and is known as the Cauchy distribution.

Let's compute the mean, which is

$$\begin{aligned} E[X] &= \frac{1}{\pi} \int_{-\infty}^{\infty} \frac{x}{1+x^2} dx \\ &= \frac{1}{\pi} \left[\int_{-\infty}^0 \frac{x}{1+x^2} dx + \int_0^{\infty} \frac{x}{1+x^2} dx \right]. \end{aligned}$$

So taking the second integral we have

$$\begin{aligned} \int_0^{\infty} \frac{x}{1+x^2} dx &= \lim_{b \rightarrow \infty} \int_0^b \frac{x}{1+x^2} dx \\ &= \lim_{b \rightarrow \infty} \frac{1}{2} \log(1+x^2) \Big|_0^b \\ &= \lim_{b \rightarrow \infty} \frac{1}{2} \log(1+b^2) = +\infty \end{aligned}$$

Oops!

Example 2. Here is an example of a density function that has a first and second moment only. Let

$$f(x) = \begin{cases} 3x^{-4} & x \geq 1 \\ 0 & \text{otherwise.} \end{cases}$$

Then the first moment exists since

$$\begin{aligned} E[X] &= \int_1^{\infty} x \frac{3}{x^4} dx \\ &= 3 \lim_{b \rightarrow \infty} \int_1^b x^{-3} dx \\ &= 3 \lim_{b \rightarrow \infty} -\frac{1}{2} x^{-2} \Big|_1^b \\ &= 3/2 \lim_{b \rightarrow \infty} 1 - \frac{1}{b^2} \\ &= 3/2. \end{aligned}$$

and the second moment exists as well since

$$\begin{aligned} E[X^2] &= \int_1^{\infty} x^2 \frac{3}{x^4} dx \\ &= 3 \lim_{b \rightarrow \infty} \int_1^b x^{-2} dx \\ &= 3 \lim_{b \rightarrow \infty} -x^{-1} \Big|_1^b \\ &= 3 \lim_{b \rightarrow \infty} 1 - \frac{1}{b} \\ &= 3. \end{aligned}$$

But

$$E[X^3] = \int_1^{\infty} x^3 \frac{3}{x^4} dx = 3 \int_1^{\infty} x^{-1} dx = \infty.$$

So what about the Moment generating function. Unfortunately the news here is even worse since

$$\begin{aligned} M(t) &= \int_{-\infty}^{\infty} e^{xt} 3x^{-4} dx \\ &= \int_1^{\infty} 3e^{xt} x^{-4} dx \end{aligned}$$

diverges (does not converge) for any $t > 0$ since

$$\lim_{x \rightarrow \infty} \frac{e^{xt}}{x^4} = +\infty.$$

Def. (Characteristic Function) Let X be a random variable. The characteristic function for X is defined as

$$\phi(t) = \begin{cases} \int_{-\infty}^{\infty} e^{itx} f(x) dx & \text{if } X \text{ is a continuous random variable} \\ \sum_{p(x)>0} e^{itx} p(x) & \text{if } X \text{ is a discrete random variable.} \end{cases}$$

Theorem. The characteristic function for a random variable X always exists and satisfies the following.

- a. $\phi(0) = 1$ and $|\phi(t)| \leq 1$ for all t .
- b. $\phi(t)$ is uniformly continuous on $\mathcal{R}$.
- c. If $E|X^k| < \infty$, then $\phi^{(k)}(0) = i^k E[X^k]$.

Example 3. Let $X \sim N(\mu, \sigma^2)$. Then $\phi(t) = \exp(i\mu t - \sigma^2 t^2/2)$.

Example 4. The characteristic function for the Cauchy distribution is

$$\phi(t) = e^{-|t|}.$$

Clearly $\phi(0) = 1$ as expected. And it not differentiable at 0, which is also as expected since $E[|X|] = \infty$.

In the case that X is a continuous random variable, you may have noticed that the characteristic function is the Fourier transform of the density function $f(x)$.

There is much more that we could say about this and it is a powerful and important tool. But I will not make further use of the characteristic function in this course.

Let $X \sim N(\mu, \Sigma)$ be an n dimensional normally distributed random variable. Then

$$f(x, \mu, \Sigma) = \left(\frac{1}{\sqrt{2\pi}} \right)^n |\Sigma|^{-1/2} \exp \left(-\frac{1}{2} (x - \mu)^T \Sigma^{-1} (x - \mu) \right).$$

Theorem. Let $X \sim N(\mu, \Sigma)$ be an n dimensional random variable. Let $Y = AX + b$, where A is m by n with $m \leq n$ and b is a column vector of length m . Then

$$Y \sim N(A\mu + b, A\Sigma A^T).$$

Important application Let $Z_1, \dots, Z_n$ be independent $N(0, 1)$ random variables. Then $Z = (Z_1, \dots, Z_n)^T \sim N(0, I)$, where I is the $n \times n$ identity matrix.

Let Σ be any covariance matrix (any matrix that is positive definite and symmetric). Then the Cholesky decomposition of Σ allows us to write

$$\Sigma = L L^T,$$

where L is lower triangulate.

If we let $X = LZ + \mu$, then $X \sim N(\mu, \Sigma = L L^T)$.

This is the standard method for generating normally distributed data with given mean μ and covariance matrix Σ .

In addition to the moment generating function and the characteristic function, another useful function is what is generally called the generating function. These are very useful for discrete random variables in particular. Suppose that X is a discrete random variable with mass points at the non-negative integers and as usual, let $P(X = i) = p_i$.

Def. The Generating Function for X is

$$E(s^X) = \sum_{n=0}^{\infty} s^n p_n,$$

since $P(X = n) = p_n$.

Example. Suppose that X is a Poisson distributed random variable with parameter λ . Then

$$G(s) = E(s^X) = \sum_{k=0}^{\infty} s^k \frac{\lambda^k}{k!} e^{-\lambda} = e^{\lambda(s-1)}.$$

Compare this to the moment generating function for a Poisson random variable which is

$$M(t) = e^{\lambda(e^t-1)}.$$

Of course - all you need is to note that $M(t) = G(s = e^t)$.